(i) When $a = 0$,

$$\int_0^b x^2\,dx = \left(\int_0^b x\,dx\right)^2$$

$$\frac{b^3}{3} = \left(\frac{b^2}{2}\right)^2$$

$\because b > 0$

$$\therefore b = \frac{4}{3}$$

(ii) When $a = 1$,

$$\int_1^b x^2\,dx = \left(\int_1^b x\,dx\right)^2$$

$$\frac{b^3 - 1}{3} = \left(\frac{b^2 - 1}{2}\right)^2$$

$$3b^4 - 4b^3 - 6b^2 + 7 = 0$$

$$(b-1)(3b^3 - b^2 - 7b - 7) = 0$$

$\because b > a = 1$

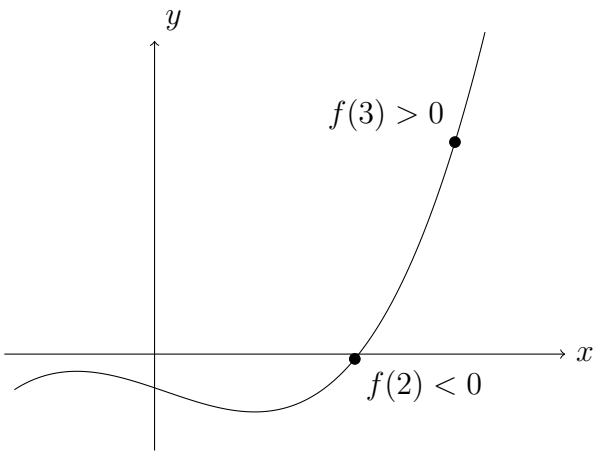
$$\therefore 3b^3 - b^2 - 7b - 7 = 0$$

Let $f(b) = 3b^3 - b^2 - 7b - 7$

$$f'(b) = 9b^2 - 2b - 7 = (b-1)(9b+7)$$

Stationary points occur at $b = -\frac{7}{9},\ 1$

$$f\left(-\frac{7}{9}\right) = -\frac{868}{243} < 0, \qquad f(1) = -12 < 0$$



The two stationary points are below the b-axis, so there is only one real root

$$f(2) = -1 < 0, \qquad f(3) = 44 > 0$$

$$\therefore 2 < b < 3$$

(iii) Let $p = b + a,\ q = b - a$

$$b^2 - a^2 = (b - a)(b + a) = pq$$

$$b^3 - a^3 = (b - a)(b^2 + ab + a^2)$$

$$b^2 + ab + a^2 = \frac{3p^2 + q^2}{4}$$

$$\int_a^b x^2\,dx = \left(\int_a^b x\,dx\right)^2$$
$$\frac{b^3 - a^3}{3} = \left(\frac{b^2 - a^2}{2}\right)^2$$
$$\frac{q(3p^2 + q^2)}{12} = \frac{p^2q^2}{4}$$

$\because q > 0$

$$\therefore 3p^2 + q^2 = 3p^2q$$

$$p^2 = \frac{q^2}{3(q - 1)}$$

$\because p = b + a \geq b - a = q,$

$$\therefore p^2 \geq q^2$$

$$\frac{q^2}{3(q - 1)} \geq q^2$$

$$\implies q \leq \frac{4}{3}$$

$p^2 = \dfrac{q^2}{3(q - 1)} > 0 \implies q > 1$

$$1 < q \leq \frac{4}{3} \implies 1 < b - a \leq \frac{4}{3}$$

This solution was submitted by **Jon**

Published and refined by **Jon**

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